

Jake F. Capmeil

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EDUCATION

Hunter College High School

New York, NY

- Relevant Coursework: Honors Chemistry, Honors Computer Science, Extended-Honors Math, Honors Biology
- Awards: National French Grand Concours (Native Speakers Exclusively), Two-Time Gold Medalist

PROGRAMS & HONORS

Carnegie Mellon University, CS Scholars Program

Summer 2026

- Fully funded, merit-based program with a ~4% acceptance rate
- Four-week residential experience studying college-level CS and mathematics through project-based learning, culminating in a research presentation at the closing symposium
- Conducted research, under the mentorship of a professor, on individuals' likelihood of turning to AI rather than human assistance as a study aid, sampling across age groups, levels of education, and academic rigor, and using Python and a custom-built machine learning algorithm to model and analyze the results

Science Research Seminar

2025 to 2026

- Highly selective research seminar preparing students for work in research labs
- Analyzed laboratory publications, passed a qualifying exam, and practiced interpreting and presenting research findings
- Covered advanced topics in scientific disciplines, research ethics, and statistics
- Completed an accompanying rigorous course in Machine Learning and Advanced Python, applying skills to hands-on ML projects

English Peer Tutor

2025 to 2026

- Selected as one of the strongest writers in my grade to provide structured writing mentorship to younger students
- Deliver targeted instruction in grammar, organization, analytical clarity, and creative composition

RESEARCH & PROJECTS

Investigation on the Effects of Ocean Bathymetry on Wave Size

Jan to May 2024

Solo Project

- Designed and constructed a functional wave tank to experimentally model coastal bathymetry
- Fabricated interchangeable underwater topographies to quantify how slope, depth, and contour influence wave height and waveform transformation
- Collected and analyzed measurements of wave propagation and shoaling effects across varying bathymetric models

The Influence of Wing Shape on Aircraft Flight

Jan to May 2025

Team Project

- Engineered a modular sailplane fuselage enabling interchange between wing designs for controlled testing
- Designed multiple wing configurations with constant planform area but varying aspect ratios to isolate effects on lift, stability, and glide efficiency

- Conducted systematic flight trials, recorded performance metrics, and analyzed results to determine the most efficient design

Testing the "Centriphone" Model with Variations on Wing Design

Dec 2024 to Feb 2026

Solo Investigative Project

- 3D-printed two wing designs for a centrifugal camera-mount system for dynamic 360° video capture while skiing, testing aerodynamic effects of different span-chord ratios at constant surface area
- Assessed effects on stability and flight path consistency, then iterated on handle geometry to optimize stability and user control following damage sustained during initial testing

ACTIVITIES & EXPERIENCE

Cross Country & Track and Field, Team Captain

2023 to Present

- Year-round varsity athlete training seven days per week, ranked among the top athletes in grade and school across all events
- Captain of both the cross country and track teams, coordinating workouts, mentoring underclassmen, and directing team communication and race-day logistics

Interact Club of Chinatown, Vice President & Co-Founder

2024 to Present

- Co-founded and lead the youth chapter of Chinatown's Rotary Club, building the club's board structure, membership, and internal communications from the ground up
- Raised \$1,500 in the club's first year and coordinated recurring volunteer events, including a community clothing drive and a Rotary Pinning Ceremony, in partnership with local organizations

Photography Assistant

2024 to Present

- Assisted on-set production across shoots and managed transport, setup, and breakdown of camera and lighting equipment
- Shot and edited promotional content
- Reviewed, selected, and organized images, then packaged and shipped fine art prints for gallery clients

Catholic Church Volunteer

2023 to Present

- Assist in biweekly refugee clothing drives, multilingual book drives, and general property maintenance

Laru Beya Summer Volunteer

Summer 2024 to Present

- Assisted with nonprofit surf instruction and water safety programming for underserved youth

SKILLS

Programming & Data: Python, data processing and analysis (pandas), machine learning algorithms (supervised learning, neural networks, data pipelines), game development

Web & Version Control: Full-stack web development (Supabase, GitHub), building and managing live sites and applications

Lab Techniques: PCR, Gel Electrophoresis, Micropipetting, Distillation

Languages: Fluent in English and French

Leadership & Coordination: Team captain and club officer with experience organizing events, board structure, communications, and volunteer programs

AI Tools: Claude, ChatGPT, and Gemini, with experience in AI-assisted coding and data analysis using Claude

Software: Excel, Word, PowerPoint, Adobe Express, Affinity